Michael Kukarkin

Curriculum Vitae

Education

2019–Present **Masters of Database Technologies**, *Saint Petersburg State University*, Saint Petersburg.

2015–2019 **Bachelor of Applied Mathematics, Fundamental Informatics and Programming**, *Saint Petersburg State University*, Saint Petersburg.
Specialized in Computer Software and Dataware

Bachelor Thesis

Title *Sentiment Analysis methods of user messages in social networks*

Supervisor Doctor Ivan Blekanov

Description This thesis explored the methods of sentiment analysis of user messages in ad-hoc discussions in social networks.

Experience

Vocational

Spring, 2020–Present **Android Kotlin Developer**, CASE, Saint Petersburg.
Single-handedly developing an Android chat application focused on business communications with Firebase as a backend.

During this time I developed the right database architecture for needs of the application, implemented necessary server-side functions: to monitor user activity and to deliver push-notifications via Firebase Cloud Messaging, etc. Also I supervised the development of the same application for iOS.
As a consequence I learned a lot about Android development and gained experience working with:

- Android Architecture Components
- Kotlin Coroutines
- Firebase infrastructure
- Room Persistence Library
- dependency injection libraries

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2018–2020 **Software Developer**, RALIENT, LLC, Saint Petersburg.

Developed and maintained control system for autonomous vehicle. Developed remote control system for electic vehicle. Implemented software for electric vehicle's actuators and localization sensors. Took part in developing company's simulator software.

Detailed achievements:

- Implemented small numerical optimization library:
  - Conjugate gradient
  - Gradient descent
- Implemented autonomous vehicle control algorithm
- Implemented smoothing algorithm for lane detection
- Implemented Kalman filtering for localization and IMU sensors
- Mastered CanOpen/serial/tcp/udp protocols

2017–2019 **Research Assistant**, School of Journalism and Mass Communication, Saint Petersburg.

Developed machine learning and statistical algorithms for sentiment analysis in social media. Wrote reports on current state of natural language processing. Wrote 3 papers on related topics.

Research Grants

2017–2019 **Reasercher**, *Presidential grant for young scholars — Doctors of Science, grant MD-6259.2018.6*.

Scientific papers

2018 Sentiment analysis for ad hoc discussions using multilingual knowledge-based approach
ACM International Conference Proceeding Series (SCOPUS cited)

2018 Negative A/Effect: Sentiment of French-speaking users and its impact upon affective hashtags on Charlie Hebdo
Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) (SCOPUS cited)

2018 Multi-dimensional echo chambers: Language and sentiment structure of Twitter discussions on the Charlie Hebdo case
Communications in Computer and Information Science (SCOPUS cited)

2019 Topics in the Russian Twitter and Relations between their Interpretability and Sentiment
2019 6th International Conference on Social Networks Analysis, Management and Security, SNAMS 2019 (SCOPUS cited)

Conference participations

2018 Oral Presentation at The 3rd International Conference on Applications in Information Technology

Hard skills

Basic JAVA, TYPESCRIPT, Gradle

Intermediate KOTLIN, C++, PYTHON, LaTeX, Linux, cmake, Kotlin Coroutines, JSON, GIT, MVVM, Firebase

Languages

Russian **Mothertongue**

English **Upper Intermediate**

Online courses

- Computer Science Center C++: Programming
- Computer Science Center: Algorithms
- Samsung research Russia: Neural networks and computer vision

Interests

- Programming
- Natural language processing
- Cooking
- Machine Learning
- Computer vision

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